

Diet for Fatty Liver Disease: Recommended Patterns, Foods to Eat, and Foods to Avoid

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Background: diet as a cornerstone of fatty liver disease management

Diet is a main part of treating fatty liver disease because it can help reduce body weight, improve insulin sensitivity, and improve liver-related markers. The strongest overall message from these studies is that sustained calorie reduction and a healthy dietary pattern, especially a Mediterranean-style pattern, matter more than any single "superfood." (7 sources)

Diet is a core part of fatty liver disease care, not just a side recommendation. Reviews of NAFLD management note that dietary interventions play a crucial role, especially diets that support weight loss and better insulin sensitivity, and a meta-analysis found a dose-response link between greater calorie restriction and better weight loss and liver outcomes, supporting calorie reduction as a key part of management. ([Kounatidis et al., 2024](#)) ([Wang et al., 2023](#)) More broadly, plant-based, high-fiber, low-fat eating has long been recommended in fatty liver disease. ([Li et al., 2021](#))

Within this broader approach, the Mediterranean diet has the most consistent support in the papers provided. Meta-analyses of randomized trials suggest it can improve markers linked to NAFLD severity, including liver enzymes, cholesterol, waist size, and possibly fibrosis, even though some outcomes vary across studies. ([Kounatidis et al., 2024](#)) ([Sangouni et al., 2021](#)) ([Bo' et al., 2023](#)) In one trial, better adherence to a Mediterranean diet was linked with greater reductions in intrahepatic fat and better metabolic health measures over 6 months. ([Montemayor et al., 2022](#)) Other healthy dietary patterns may also help; for example, a randomized trial found that a Paleolithic diet reduced liver fat and improved insulin sensitivity-related measures in people with type 2 diabetes. ([Kounatidis et al., 2024](#)) ([Otten et al., 2018](#)) Overall, the practical takeaway is that fatty liver disease usually improves most when patients adopt a sustainable eating pattern that lowers calorie intake, supports weight loss, and is built around minimally processed, fiber-rich foods rather than highly processed foods. ([Kounatidis et al., 2024](#)) ([Li et al., 2021](#))

Evidence

([Kounatidis et al., 2024](#))

[NAFLD in the 21st Century: Current Knowledge Regarding Its Pathogenesis, Diagnosis and Therapeutics](#)

"Dietary interventions play a crucial role in managing NAFLD. A diet that promotes weight loss and improves insulin sensitivity is often recommended. In a recent metaanalysis of various interventions in NAFLD, there was a dose-response association between the degree of calorie restriction and the favorable effects on weight loss and liver function, advocating that this approach remains the key component of NAFLD management (Wang et al., 2023)"

".Metaanalyses of randomized control trials (RCTs) have shown that the Mediterranean diet may reduce indirect and direct outcomes associated with NAFLD severity, such as liver fibrosis, total cholesterol, waist circumference, and liver enzymes (Bo' et al.,

2023)(Sangouni et al., 2021)"

"Regarding nutrition, the Mediterranean diet and the Paleolithic diet have been demonstrated to exert beneficial effects (Montemayor et al., 2022)[79](Otten et al., 2018). The Mediterranean diet is rich in fruits and vegetables, allows moderate consumption of fish, and avoids red meat."

(Wang et al., 2023)

Effects of various interventions on non-alcoholic fatty liver disease (NAFLD): A systematic review and network meta-analysis

"Background: With the increasing prevalence of obesity and metabolic syndrome, the incidence of non-alcoholic fatty liver disease (NAFLD) is also increasing. In the next decade, NAFLD may become the main cause of liver transplantation. Therefore, the choice of treatment plan is particularly important. The purpose of this study was to compare several interventions in the treatment of NAFLD to provide some reference for clinicians in selecting treatment methods. Methods: We searched Public Medicine (PubMed), Medline, Excerpta Medica Database (Embase), and Cochrane Library from January 2013 to January 2023 to identify randomized controlled trials (RCTs) published in English. The network meta-analysis was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Forty-three studies accounting for a total of 2,969 patients were included, and alanine aminotransferase (ALT), total cholesterol (TC), and low-density lipoprotein cholesterol (LDL) were selected as outcome measures for analysis and comparison. Results: We evaluated the results of drug, diet, and lifestyle interventions between the intervention and control groups. Curcumin (CUN) and probiotics (PTC) were selected for medication, the Mediterranean diet (MDED) was selected for special diet (SPD), and various kinds of exercise and lifestyle advice were selected for lifestyle interventions (LFT). The SUCRA was used to rank interventions according to the effect on ALT indicators (SUCRA: PTC 80.3%, SPD 65.2%, LFT 61.4%, PLB 32.8%, CUN 10.2%), TC indicators (SUCRA: PTC 89.4%, SPD 64%, CUN 34%, LFT 36.6%, PLB 17%), and LDL indicators (SUCRA: PTC 84.2%, CUN 69.5%, LFT 51.7%, PLB 30.1%, SPD 14.5%). The pairwise meta-analysis results showed that MDED was significantly better than NT in improving ALT [SMD 1.99, 95% CI (0.38, 3.60)]. In terms of improving TC and LDL, ATS was significantly better than NT [SMD 0.19, 95% CI (0.03, 0.36)] [SMD 0.18, 95% CI (0.01, 0.35)]. Conclusion: Our study showed that PTC is most likely to be the most effective treatment for improving NAFLD indicators. Professional advice on diet or exercise was more effective in treating NAFLD than no intervention."

(Li et al., 2021)

Plant-Based Foods and Their Bioactive Compounds on Fatty Liver Disease: Effects, Mechanisms, and Clinical Application

"The plant-based, high-fiber, and low-fat diet has been recommended to manage FLD for a long time."

(Sangouni et al., 2021)

Effect of Mediterranean diet on liver enzymes: a systematic review and meta-analysis of randomised controlled trials

"Abstract Elevated levels of liver enzymes are the main markers of liver dysfunction. Liver enzymes are the important indicators of non-alcoholic fatty liver disease (NAFLD) in the general population. Previous randomised clinical trials (RCT) investigated the effects of Mediterranean diet (MedDiet) as a plant-based diet on features of NAFLD like liver enzymes, but their results are contradictory. This study aimed to systematically review and meta-analyse RCT investigating the effect of MedDiet on liver enzymes. PubMed, Web of Science, Scopus and Google Scholar were searched until December 2020. A total of ten RCT (n 705 participants) evaluating the effect of MedDiet on liver enzymes including aspartate aminotransferase (AST), alanine transaminase (ALT) and γ -glutamyltransferase (GGT) were included. A random effects model was used to estimate the pooled effect size. To evaluate the heterogeneity among the included studies, the Cochran's Q-test and I-squared test were used. The MedDiet significantly reduced AST (weighted mean difference (WMD) = -0.38 IU/l; 95% CI - 0.73, -0.03 IU/l; P = 0.03) and GGT (WMD = -0.16 IU/l; 95% CI - 0.32, -0.006 IU/l; P = 0.04) but had no significant effect on ALT (WMD = -0.55 IU/l; 95% CI - 1.25, 0.13 IU/l; P = 0.11). However, sensitivity analysis revealed that the overall effects of MedDiet on AST, GGT and ALT were significantly influenced by removing some studies. There was no publication bias based on Begg's and Egger's tests. Generally, MedDiet can improve liver enzymes. To better conclusion, further RCT investigating the effect of MedDiet on liver enzymes, especially in patients with NAFLD, are still required."

(Bo' et al., 2023)

Does the Mediterranean Diet Have Any Effect on Lipid Profile, Central Obesity and Liver Enzymes in Non-Alcoholic Fatty Liver Disease (NAFLD) Subjects? A Systematic Review and Meta-Analysis of Randomized Control Trials

"The effectiveness of the Mediterranean diet (MD) in non-alcoholic fatty liver disease (NAFLD) subjects has been evaluated in several randomized controlled trials (RCTs). This systematic review and meta-analysis aimed to evaluate the overall effects of MD intervention in a cohort of NAFLD patients targeting specific markers such as central obesity, lipid profile, liver enzymes and fibrosis, and intrahepatic fat (IHF). Google Scholar, PubMed, and Scopus were explored to collect relevant studies from the last 10 years. RCTs with NAFLD subjects were included in this systematic review with a mean intervention duration from 6 weeks to 1 year, and different intervention strategies, mainly including energy restriction MD (normal or low glycaemic index), low-fat MD with increased monounsaturated and polyunsaturated fatty acids, and increased exercise expenditure. The outcomes measured in this meta-analysis were gamma-glutamyl transferase (GGT), alanine aminotransferase (ALT), total cholesterol (TC), waist circumference (WC), and liver fibrosis. Ten randomized controlled trials, which involved a total of 737 adults with NAFLD, were included. According to the results, the MD seems to decrease the liver stiffness (kPa) by -0.42 (CI95% -0.92, 0.09) (p = 0.10) and significantly reduce the TC

by -0.46 mg/dl (CI95% -0.55 , -0.38) ($p = 0.001$), while no significant findings were documented for liver enzymes and WC among patients with NAFLD. In conclusion, the MD might reduce indirect and direct outcomes linked with NAFLD severity, such as TC, liver fibrosis, and WC, although it is important to consider the variations across trials. Further RCTs are necessary to corroborate the findings obtained and provide further evidence on the role of the MD in the modulation of other disorders related to NAFLD."

(Montemayor et al., 2022)

Adherence to Mediterranean Diet and NAFLD in Patients with Metabolic Syndrome: The FLIPAN Study

"Unhealthy diet is an important factor in the progression of non-alcoholic fatty liver disease (NAFLD). Previous studies showed the benefits of a Mediterranean diet (MedDiet) on Metabolic syndrome (MetS), type 2 diabetes mellitus (T2DM), and cardiovascular diseases, which usually have a pathophysiological relationship with NAFLD. To assess the effect of adherence to a MedDiet on NAFLD in MetS patients after lifestyle intervention, this multicentre (Mallorca and Navarra, Spain) prospective randomized trial, with personalized nutritional intervention based on a customized MedDiet, coupled with physical activity promotion was performed to prevent, and reverse NAFLD among patients with MetS. The current analysis included 138 patients aged 40 to 60 years old, Body Mass Index (BMI) $27-40$ kg/m², diagnosed with NAFLD using MRI, and MetS according to the International Diabetes Federation (IDF). A validated food frequency questionnaire was used to assess dietary intake. Adherence to Mediterranean diet by means of a 17-item validated questionnaire, anthropometrics, physical activity, blood pressure, blood biochemical parameters, and intrahepatic fat contents (IFC) were measured. The independent variable used was changes in MedDiet adherence, categorized in tertiles after 6 months follow-up. Subjects with high adherence to the MedDiet showed higher decreases in BMI, body weight, WC, SBP, DBP, and IFC. An association between improvement in adherence to the MedDiet and amelioration of IFC after 6-month follow-up was observed. High adherence to the MedDiet is associated with better status of MetS features, and better values of IFC."

(Otten et al., 2018)

A heterogeneous response of liver and skeletal muscle fat to the combination of a Paleolithic diet and exercise in obese individuals with type 2 diabetes: a randomised controlled trial

"Aims/hypothesis The aim of the study was to investigate ectopic fat deposition and insulin sensitivity, in a parallel single-blinded randomised controlled trial, comparing Paleolithic diet alone with the combination of Paleolithic diet and exercise in individuals with type 2 diabetes. Methods Thirty-two individuals with type 2 diabetes with BMI $25-40$ kg/m² and $30-70$ years of age followed a Paleolithic diet ad libitum for 12 weeks. In addition, study participants were randomised by computer program to either supervised combined exercise training (PD-EX group) or standard care exercise recommendations (PD group). Staff performing examinations and assessing outcomes were blinded to group assignment. Thirteen participants were analysed in each group: hepatic and peripheral insulin sensitivity were measured using the hyperinsulinaemic-euglycaemic clamp technique combined with [6,6-²H₂]glucose infusion, and liver fat was assessed by proton magnetic resonance spectroscopy; both analyses were secondary endpoints. Intramyocellular lipid (IMCL) content was measured by magnetic resonance spectroscopy as a secondary analysis. All examinations were performed at Umeå University Hospital, Umeå, Sweden. Results Both study groups showed a median body weight loss of 7 kg. Fat mass decreased by 5.7 kg in the PD group and by 6.5 kg in the PD-EX group. Maximum oxygen uptake increased in the PD-EX group only. Liver fat showed a consistent reduction (74% decrease) in the PD group, while the response in the PD-EX group was heterogeneous ($p < 0.05$ for the difference between groups). IMCL content of the soleus muscle decreased by 40% in the PD group and by 22% in the PD-EX group ($p < 0.05$ for the difference between groups). Both groups improved their peripheral and adipose tissue insulin sensitivity, but not their hepatic insulin sensitivity. Plasma fetuin-A decreased by 11% in the PD group ($p < 0.05$) and remained unchanged in the PD-EX group. Liver fat changes during the intervention were correlated with changes in fetuin-A ($r_s = 0.63$, $p < 0.01$). Participants did not report any important adverse events caused by the intervention. Conclusions/interpretation A Paleolithic diet reduced liver fat and IMCL content, while there was a tissue-specific heterogeneous response to added exercise training. Trial registration ClinicalTrials.gov NCT01513798 Funding Swedish Diabetes Research Foundation, County Council of Västerbotten, Swedish Heart and Lung Foundation, King Gustav V and Queen Victoria's Foundation"

Recommended dietary pattern for fatty liver disease

The most consistently recommended eating pattern for fatty liver disease is a calorie-controlled Mediterranean-style diet. In practice, that means mostly plant foods and minimally processed foods, with healthy fats like olive oil and nuts, regular fish, and much less sugar, refined carbs, and saturated fat. (24 sources)

Across the papers provided, the clearest dietary recommendation is a Mediterranean-style pattern, and several reviews note that it is the diet most often recommended by clinical guidelines for NAFLD management. (Eslamparast et al., 2017) (Riazi et al., 2019) (Fougerat et al., 2020) (Godoy-Matos et al., 2020) (Mantovani et al., 2021) (Lujan et al., 2021) (Paternostro et al., 2022) (Monserrat-Mesquida et al., 2023) (Espina et al., 2023) (Sun et al., 2024) This pattern overlaps with what some authors call a "high-quality healthy diet": high in vegetables, fruits, legumes, whole grains, nuts, seeds, and fiber; built around olive oil and other sources of monounsaturated and omega-3 fats; and lower in red meat, processed meat, sweets, refined carbohydrates, and highly processed foods. (Oca et al., 2020) (Riazi et al., 2019) (Sebastiani et al., 2021)

[\(Patel et al., 2022\)](#) [\(Lucaciu et al., 2025\)](#)

A practical way to think about the recommended pattern is: make most meals plant-forward, use olive oil instead of butter or other saturated fats, choose fish or poultry more often than red or processed meat, and keep added sugars and refined starches low. This matches the recurring findings that many patients with fatty liver disease tend to overconsume simple carbohydrates and saturated fat while under-consuming fiber and omega-3-rich foods, and that shifting in the opposite direction is helpful. [\(Eslamparast et al., 2017\)](#) [\(Freidoony et al., 2014\)](#) [\(Ristic-Medic et al., 2020\)](#) [\(Martin et al., 2022\)](#) [\(Kichloo et al., 2023\)](#) Some reviews also describe plant-based patterns such as DASH-like or Mediterranean-related diets as useful because they lower saturated fat, red and processed meat, and refined carbohydrate intake while increasing antioxidant- and fiber-rich foods. [\(Monserrat-Mesquida et al., 2023\)](#) [\(Barraza et al., 2024\)](#)

The Mediterranean pattern is also usually recommended together with energy control rather than as a license to eat unlimited “healthy” foods. Several papers pair this diet with modest calorie restriction to support gradual weight loss, often aiming for about 0.5 to 1 kg per week, and discourage extreme very-low-calorie approaches because they are hard to sustain. [\(Godoy-Matos et al., 2020\)](#) [\(Brunner et al., 2019\)](#) [\(Ratziu et al., 2019\)](#) [\(Ristic-Medic et al., 2020\)](#) [\(Dyson et al., 2014\)](#) [\(Dyson et al., 2014\)](#) So, the recommended diet for most patients is not a single rigid menu, but a sustainable, calorie-aware Mediterranean-style eating pattern centered on minimally processed plant foods, healthy unsaturated fats, and lean protein sources. [\(Pickett-Blakely et al., 2018\)](#) [\(Godoy-Matos et al., 2020\)](#) [\(Paternostro et al., 2022\)](#)

Evidence

(Eslamparast et al., 2017)

[Dietary Composition Independent of Weight Loss in the Management of Non-Alcoholic Fatty Liver Disease](#)

"The majority of NAFLD patients follow diets with overconsumption of simple carbohydrates, total and saturated fat, with reduced intake of dietary fiber and omega-3 rich foods"

".we concur with the EASL-EASD-EASO Clinical Guidelines recommendation of the Mediterranean diet as the diet of choice in these patients."

(Riazi et al., 2019)

[Dietary Patterns and Components in Nonalcoholic Fatty Liver Disease \(NAFLD\): What Key Messages Can Health Care Providers Offer?](#)

"The Mediterranean (MD) diet has been recommended as the diet of choice for the treatment of NAFLD by various clinical practice guidelines [92]"

"In the MD diet, high consumption of extra-virgin olive oil, vegetables (especially root and green varieties), fruits (particularly fresh), whole grains (cereals, breads, rice, or pasta), legumes, nuts (walnuts, hazelnuts, or almonds), and moderate consumption of fish (especially fatty fish rich in omega-3 fatty acids), low-fat dairy products, and red wine is recommended over the consumption of red and processed meat, processed and high-sugar food, refined carbohydrates and high-fat milk products [83]."

(Fougerat et al., 2020)

[Peroxisome Proliferator-Activated Receptors and Their Novel Ligands as Candidates for the Treatment of Non-Alcoholic Fatty Liver Disease](#)

"Currently, the Mediterranean diet is recommended for the management of NAFLD (3440607)"

". Given the detrimental hepatic effects of carbohydrates, especially fructose as described above, very low-carbohydrate ketogenic diets have received attention for the management of NAFLD."

(Godoy-Matos et al., 2020)

[NAFLD as a continuum: from obesity to metabolic syndrome and diabetes](#)

"The dietary management in NAFLD should focus on caloric restriction, aiming to induce 0.5-1 kg/week of weight loss. In general, a low-calorie diet should have 50-60% of caloric intake from carbohydrates and 20-25% from lipids (Brunner et al., 2019)(Ratziu et al., 2019). Saturated fats should account for < 10% of total fat intake (Brunner et al., 2019)(Ratziu et al., 2019)"

"Therefore, snacking should be avoided in patients with NAFLD (Ratzu et al., 2019)"

"The Mediterranean diet, which is rich in monounsaturated fatty acids, PUFA and polyphenols, showed improvement on insulin sensitivity and hepatic steatosis. According to many medical societies, Mediterranean diet is an eating pattern of choice in individuals with NAFLD (3440607)(Aller et al., 2015)(Kontogianni et al., 2014)(Ryan et al., 2013)(Misciagna et al., 2016)"

"Very low-calorie diets (500-800 kcal/day) have low long-term adherence and are not recommended (Brunner et al., 2019)(Ratzu et al., 2019)."

(Mantovani et al., 2021)

Treatments for NAFLD: State of Art

"All patients with NAFLD, regardless of the presence of T2DM, should avoid alcohol consumption (even moderate) and, when possible, the use of hepatotoxic drugs [1,2]. Clinicians should also recommend avoiding cigarette smoking [1,2,(Glen et al., 2016) (Charatcharoenwitthaya et al., 2020) and fructose-containing beverages and foods [1,2,(Glen et al., 2016)(Vos et al., 2013)"

"at present, the Mediterranean diet is the most recommended dietary pattern in NAFLD patients by most guidelines [1,2,(Zelber-Sagi et al., 2017). In brief, the Mediterranean diet is characterized by a high intake of vegetables, legumes, whole grains, olive oil (as the main source of added fat), fish, seafood, nuts, fruits, as well as a low intake of red meat, processed meats, and sweets [18,22]26]."

(Lujan et al., 2021)

Overview of Non-Alcoholic Fatty Liver Disease (NAFLD) and the Role of Sugary Food Consumption and Other Dietary Components in Its Development

"Most current guidelines recommend low-calorie, plant-based diets, such as the Mediterranean diet, as the most effective dietary pattern to treat NAFLD"

"general dietetic recommendations for these patients must include moderate weight loss (5%), maintenance of a low-calorie diet, cessation of alcohol and tobacco consumption, low carbohydrate intake ($\leq 40\%$ of daily energy intake), decreased fat (especially SFAs and TFAs), and simple carbohydrate consumption [69]."

(Paternostro et al., 2022)

Current treatment of non-alcoholic fatty liver disease

"Energy restriction and exclusion of NAFLD-promoting components (i.e. processed food, products high in added fructose) are recommended by the EASL-NAFLD guidelines [5] and generally speaking a "Mediterranean diet" should be recommended to all NAFLD patients [5]."

(Montserrat-Mesquida et al., 2023)

Plasma Fatty Acid Composition, Oxidative and Inflammatory Status, and Adherence to the Mediterranean Diet of Patients with Non-Alcoholic Fatty Liver Disease

"there is strong evidence that the Mediterranean diet (MedDiet) is recommended as a main treatment for the management of NAFLD (Bischoff et al., 2020). Plant-based diets, such as the Dietary Approaches to Stop Hypertension (DASH) and the MedDiet, with recommendations like low intake of saturated fatty acids (SFAs), red and processed meats, and refined carbohydrates (CHOs) are beneficial for NAFLD (Hydes et al., 2020)"

"The consumption of ultra-processed foods (UPFs), which are industrial formulations manufactured with poor nutritional composition, is harmful to the health, because it induces high levels of NAFLD-related biomarkers and increased age-related accumulation of visceral and total adiposity (Konieczna et al., 2022)(Konieczna et al., 2021)."

(Espina et al., 2023)

Evaluation and Management of Nutritional Consequences of Chronic Liver Diseases

"the Mediterranean diet (MD) is, without a doubt, the dietary pattern recommended among different scientific societies [153]. The MD diet, based on a high intake of fruits, vegetables, legumes, omega 3 PUFAs, and fish with a low intake of red meat and sweets"

"clinicians tend to recommend that it should be avoided (Figure 4)"

"there is strong evidence for the undeniable association between added sugars in fo drinks and the development of MAFLD."

(Sun et al., 2024)

Current strategies for nonalcoholic fatty liver disease treatment (Review)

"patients with NAFLd are advised to consume the Mediterranean diet in the European Association for the Study of the Liver, European Association for the Study of diabetes and European Association for the Study of Obesity clinical Practice Guidelines for

the management of NAFLD (18). The Mediterranean diet, a nutritional model, began in the regions surrounding the Mediterranean sea and consisted of eating nuts, legumes, fruits, vegetables, fish, white meat and olive oil, which are rich in monounsaturated fatty acids (MUFAs) and polyunsaturated fatty acids (PUFAs). The diet also recommends drinking red wine in moderation, and eating limited amounts of red meat, processed meats and sweets. Minimizing the intake of processed and high-fructose food is a feature of the Mediterranean diet"

"It is well established that the risk of hepatic steatosis increases after consuming high-calorie, high-sugar and high-fat foods (Koopman et al., 2014)."

(Oca et al., 2020)

Microbiota, Fiber, and NAFLD: Is There Any Connection?

"A "high-quality healthy diet" has been proposed to improve hepatic steatosis and metabolic dysfunction in patients with NAFLD, independent of caloric restriction and weight loss (Eslamparast et al., 2017). This nutritional program is supported by the idea that lower fiber intake is common in NAFLD (Eslamparast et al., 2017)(Wehmeyer et al., 2016) and is based on moderate to high carbohydrates intake (45-65% of total daily calories), low to moderate fat intake (below 30-35% of total calories with a high preference for healthy fat intake-monounsaturated fatty acids and omega-3 polyunsaturated fatty acids), protein intake (15-20% of total daily calories) and fiber intake increasing the consumption of fruits and vegetables, with more focus on prebiotic fiber. However, a high, fructose-rich diet in the form of added sugar is associated with more intestinal permeability, endotoxemia, higher hepatic TNF- α production, and lipid peroxidation, promoting hepatic steatosis and NAFLD (Romero-Gomez et al., 2017). This diet overlaps with the Mediterranean Diet (MD), which is the most recommended dietary pattern in NAFLD."

(Sebastiani et al., 2021)

Current considerations for clinical management and care of non-alcoholic fatty liver disease: Insights from the 1st International Workshop of the Canadian NASH Network (CanNASH)

"One way to achieve this, which can be even better than simple calorie restriction, is the adoption of the Mediterranean diet-a diet that is high in plant-based foods, uses olive oil as the main source of added fat, and has moderate intakes of fish, seafood, eggs, white meat, and dairy products while including limited intakes of red/processed meats and sweets, and a moderate intake of red wine (Romero-Gomez et al., 2017)."

(Patel et al., 2022)

Nonalcoholic Fatty Liver Disease in Lean/Nonobese and Obese Individuals: A Comprehensive Review on Prevalence, Pathogenesis, Clinical Outcomes, and Treatment

"the Mediterranean diet has been the most studied, and adherence to the diet has shown a reduction in the risk and progression of NAFLD via antioxidant and anti-inflammatory mechanisms linked to the consumption of monounsaturated and omega-3-fatty acids, and phytosterols found in the diet. (Saeed et al., 2019)(Daniele et al., 2016) The Mediterranean diet consists of fresh vegetables, legumes, fruit, minimally processed whole grains, fish, and foods rich in omega-3-fatty acids, such as olive oil, nuts, and seeds. It includes minimal to low consumption of dairy and red or processed meats. (Tosti et al., 2018) The EASL-EASD-EASO clinical practice guidelines recommend the Mediterranean diet for NAFLD patients. (3440607)"

"NAFLD/NASH and certain MetS conditions, such as insulin resistance, are associated with an increased intake of fructose and high fructose corn syrup products. (Ouyang et al., 2008) In patients with NAFLD, higher amounts of fructose intake via sugarsweetened beverages are associated with increased hepatic fibrosis, inflammation, and hepatocellular ballooning. (Abdelmalek et al., 2010) On the other hand, fructose found in fruits was not associated with NAFLD."

(Lucaciu et al., 2025)

Pathogenesis and Clinical Management of Metabolic Dysfunction-Associated Steatotic Liver Disease

"Different diets were proposed, but the most accepted one is the Mediterranean diet. The Mediterranean diet consists of a low consumption of sugars and refined carbohydrates and a high consumption of vegetables, fruits, whole grains, seeds, and nuts. Fish is allowed, but red meat is not recommended. It is basically a plant-based diet"

".Changing dietary habits highlights the significant role of diet in the treatment of liver disease. A hypocaloric, low-fat and low-carbohydrate diet is recommended (Guo et al., 2022)"

".Patients with MASLD should not drink alcohol. Although there is no clear scientific evidence to indicate the prohibition of alcohol consumption in MASLD, it is known that alcohol and metabolic dysfunction are synergic involved in inducing liver disease (Åberg et al., 2022)."

(Freidoony et al., 2014)

Practical approaches to the nutritional management of nonalcoholic fatty liver disease

"The Mediterranean diet which is rich in olive oil, nuts, fruits, vegetables, fish, legumes, dairy products, and wine, is an example of

food-oriented nutritional recommendation"

"Generally, hypercaloric diets, especially rich in trans/saturated fat and cholesterol, high consumption of red and processed meat, and fructose-sweetened beverages seem to increase the risk of progression toward nonalcoholic steatohepatitis (NASH), whereas reducing caloric intake and high-glycemic index (GI) foods, increasing consumption of monounsaturated fatty acids, omega-3 fatty acids, fibers, and specific protein sources such as fish and poultry have preventive and therapeutic effects."

(Ristic-Medic et al., 2020)

Calorie-Restricted Mediterranean and Low-Fat Diets Affect Fatty Acid Status in Individuals with Nonalcoholic Fatty Liver Disease

"The general recommendations for the diet are individualized, with a caloric deficit of 500–1000 kcal/day depending on the BMI of the NAFLD patient (Chalasani et al., 2018). A reduced amount of total fats should constitute less than 30% of the total energy input, with a decrease in saturated fats and refined sugar intake and an increase in soluble fiber intake (low-fat diet). New evidence supports the idea that the Mediterranean (Med diet) may have an important role in the prevention and treatment of NAFLD [9](Torres et al., 2019)(Abenavoli et al., 2014)."

(Martin et al., 2022)

Management of Dyslipidemia in Patients with Non-Alcoholic Fatty Liver Disease

"Therefore, a healthy diet (e.g., low in saturated fat with a focus on whole grain products, vegetables, fruit, and marine fish) is recommended as a cornerstone of CVD prevention in all NAFLD individuals with dyslipidemia.

The macronutrient composition should be adjusted according to a Mediterranean diet (MD) or a similar dietary pattern (2020). MD is traditionally plant based (whole grains, legumes, fruit, vegetables), low in carbohydrates (limited simple sugars and refined carbohydrates), and rich in monounsaturated (mostly olive oil) and omega-3 fatty acids, and incorporates limited amounts of red meat and low-fat dairy"

". Several dietary habits such as a high caloric diet, excess of (saturated) fat, sugar-sweetened beverages, high fructose intake, and refined carbohydrates are associated with weight gain, NAFLD, and dyslipidemia [4,38,57]. Especially high-fructose consumption has adverse effects on metabolism leading to the development of intrahepatic insulin resistance, intrahepatic fat accumulation, and the aggravation of an atherogenic lipid profile [58,59].

Daily alcohol intake should be drastically reduced (< 10 g/day for men and women). In patients with hypertriglyceridemia or advanced fibrosis (≥ F2), alcohol consumption should be completely avoided [40]."

(Kichloo et al., 2023)

Cirrhosis: Primary care approaches to screening, immunization, and lifestyle modifications

"A Mediterranean diet and avoidance of red and processed meat, high-fructose corn syrup, and foods high in saturated fats have shown the most promising results in reducing the risk of nonalcoholic fatty liver disease (NAFLD) and preventing progression to nonalcoholic steatohepatitis and eventual cirrhosis."

(Barraza et al., 2024)

Dietary Pattern's Role in Hepatic Epigenetic and Dietary Recommendations for the Prevention of NAFLD

"Alternatively, a plant-based Mediterranean-related diet is widely recommended for treating and preventing NAFLD due to its high antioxidant and anti-inflammatory properties. This diet emphasizes the consumption of fruits, vegetables, whole grains, legumes, nuts, and seeds while minimizing sugar intake (Kouvari et al., 2023). Additionally, it includes moderate fish consumption, limits red meat, and advises the exclusion of processed foods and beverages high in added fructose (George et al., 2018)"

". Adopt a diet mainly plant-based and minimize saturated fats to reduce inflammation: Consume a diet that is mainly composed of plant-based food minimizing animal-derived nutrient sources rich in saturated fats, particularly red meat."

(Brunner et al., 2019)

Nonalcoholic Fatty Liver Disease and Obesity Treatment

"Purpose of Review Nonalcoholic fatty liver disease (NAFLD), the most prevalent cause of chronic liver disease worldwide, is strongly associated with obesity and insulin resistance. Recent Findings Significant weight loss can improve NAFLD and nonalcoholic steatohepatitis (NASH). Diet and exercise that result in a sustained body weight reduction of 7–10% can improve liver fat content, NASH, and fibrosis. Vitamin E can be considered in patients with biopsy-proven NASH without diabetes, though caution must be used in those with prostate cancer. Pioglitazone improves liver histology, including fibrosis, and can be considered in patients with or without diabetes. Glucagon-like peptide-1 (GLP-1) antagonists may be beneficial in NASH, but more studies are needed before they can be recommended. Bariatric surgery, with resultant weight loss, can result in improvement in liver fat and inflammation. Summary NAFLD treatment includes diet and exercise with a target 7–10% weight reduction. Treatment goals include improvements in liver fat content, liver inflammation, and fibrosis."

(Ratzu et al., 2019)

Recommendations for Management and Treatment of Nonalcoholic Steatohepatitis

"Abstract The prevalence of nonalcoholic liver disease (NAFLD) is increasing worldwide in conjunction with the epidemic increase in obesity and metabolic risk factors. Consequently, NAFLD has become a leading indication for liver transplantation. Although genetic factors play an important role in the pathogenesis of NAFLD, detrimental lifestyle trends favoring a calorically unrestricted diet rich in carbohydrates and unsaturated fat, prolonged sedentary periods or limited physical activity have major metabolic implications. In aggregate these physiological dysregulations constitute the main risk factors for the metabolic syndrome and NAFLD. The cornerstone of the treatment of NAFLD, is lifestyle changes, including modifications to diet and physical activity, to reduce body weight and liver fat, however adherence is notoriously poor and the epidemic of NAFLD continues to grow unimpeded. In the face of this unmet clinical need, the pharmacologic therapy of NAFLD has been expanding as the varied mechanistic pathways of NAFLD are elucidated. Beyond these approaches to treating NAFLD, the prevention of other liver diseases is additionally important. Chief among these is alcoholic liver disease, and heavy use is detrimental irrespective of underlying NAFLD. However, the impact of mild to moderate alcohol use in patients with mild or nonadvanced forms NAFLD is undefined. This article summarizes the results of the International Liver Transplantation Society consensus meeting on NAFLD in liver transplantation. It describes the available evidence and provides consensus guidance on the lifestyle and pharmacologic therapies of NAFLD, and the consensus position on alcohol use in patients with NAFLD."

(Dyson et al., 2014)

Republished: Non-alcoholic fatty liver disease: a practical approach to treatment

"Until further evidence is available, a calorie restricted diet (600 Kcal less than a person needs to remain at the same weight) should be recommended aiming to lose 0.5-1 kg per week until the target weight is achieved. 11 Patients with NAFLD should be advised to avoid saturated fats, simple carbohydrates and sweetened drinks. 25 26"

"A Mediterranean diet (high in monounsaturated fatty acids), as compared with a diet low in fat and high in carbohydrates, has been shown to reduce hepatic steatosis and improve insulin sensitivity in non-diabetic subjects with NAFLD. (Ryan et al., 2013)"

(Dyson et al., 2014)

Non-alcoholic fatty liver disease: a practical approach to treatment

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(Pickett-Blakely et al., 2018)

Micronutrients in Nonalcoholic Fatty Liver Disease Pathogenesis

"For now, it is our view that the best advice for NAFLD patients is to consume a nutritionally balanced but relatively energy-restricted diet that avoids exposure to processed foods and fructosecontaining beverages, 132 akin to a sodium-reduced version of a Mediterranean-style diet, which is high in fiber, rich in anti-oxidants, and improves BMI, insulin resistance, and hepatic steatosis. (Blond et al., 2017)(Romero-Gomez et al., 2017)(Zelber-Sagi et al., 2017)(Eslamparast et al., 2017)"

Foods and food groups to emphasize

Build meals around whole, minimally processed foods: vegetables, fruits, legumes, whole grains, nuts, seeds, olive oil, and fish. Choose lean or plant proteins, get more fiber and unsaturated fats, and use these foods to replace sugary, refined, and heavily processed foods. (17 sources)

• Vegetables and fruits

- Aim to eat vegetables and fruit often, especially as the base of meals and snacks. These foods are repeatedly recommended in NAFLD because they add fiber and antioxidant compounds, and several reviews describe Mediterranean-style eating as high in vegetables, fruits, and other plant foods.

[\(Kargulewicz et al., 2014\)](#) [\(Roeb et al., 2021\)](#) [\(Lv et al., 2024\)](#)

- Deeply colored fruits and vegetables are also specifically encouraged in newer dietary summaries. [\(Zeng et al., 2024\)](#)

- **Whole grains and other high-fiber carbohydrates**
- Prefer whole grains, unprocessed cereals, legumes, and other low-glycemic, high-fiber carbohydrate sources over refined starches. Reviews recommend increasing complex carbohydrates rich in dietary fiber and choosing whole grains as a main carbohydrate source. ([Kargulewicz et al., 2014](#)) ([Perdomo et al., 2020](#)) ([Perdomo et al., 2019](#)) ([Cinque et al., 2022](#))
- This fits with broader advice to substitute ultra-processed foods with fiber-rich foods such as whole grains, vegetables, fruits, legumes, nuts, and seeds. ([Roeb et al., 2021](#))
- **Legumes, nuts, and seeds**
- Beans, lentils, chickpeas, nuts, and seeds fit the recommended pattern because they provide fiber, unsaturated fat, and often plant protein. These foods are named repeatedly as key Mediterranean-style staples for patients with fatty liver disease. ([Roeb et al., 2021](#)) ([Ferri et al., 2023](#)) ([Lv et al., 2024](#))
- Nuts are also mentioned as a useful addition in some reviews of human trial data. ([Tsamos et al., 2023](#))
- **Olive oil and other unsaturated fats**
- Use olive oil and other sources of monounsaturated and polyunsaturated fats in place of saturated fats. Multiple papers recommend increasing unsaturated fats, especially MUFAs and omega-3 fats, and describe olive oil as a main fat source in the preferred dietary pattern. ([Kargulewicz et al., 2014](#)) ([Perdomo et al., 2020](#)) ([Perdomo et al., 2019](#)) ([Kasper et al., 2020](#)) ([Cinque et al., 2022](#)) ([Lv et al., 2024](#))
- Some papers also suggest choosing cooking oils with a better fatty acid profile, such as olive or canola oil. ([Ahmed, 2015](#))
- **Fish and seafood, especially oily fish**
- Fish and seafood are commonly recommended protein choices, especially oily fish because of their omega-3 content. Reviews support diets rich in long-chain omega-3 fats and specifically encourage fish intake. ([Perdomo et al., 2020](#)) ([Perdomo et al., 2019](#)) ([Kasper et al., 2020](#)) ([Roeb et al., 2021](#)) ([Singh et al., 2017](#)) ([Lv et al., 2024](#))
- Fish oil and oily fish are often discussed as practical ways to raise omega-3 intake. ([Singh et al., 2017](#)) ([Tsamos et al., 2023](#))
- **Lean and plant-based proteins**
- Choose plant proteins often, and when eating animal protein, favor fish, seafood, eggs, poultry, and other lean options. Several reviews recommend vegetable protein, white meat, and lean proteins while minimizing fattier meat choices. ([Perdomo et al., 2020](#)) ([Perdomo et al., 2019](#)) ([Kargulewicz et al., 2014](#)) ([Rinaldi et al., 2021](#)) ([Sharma et al., 2023](#))
- Low-fat or fat-free dairy products are also included in some recent recommendations. ([Zeng et al., 2024](#))
- **Fiber-rich and gut-friendly foods**
- High-fiber intake is a recurring theme across the papers. In practice, this means emphasizing vegetables, fruit, legumes, and whole grains, and in some reviews also considering prebiotic fiber and probiotic-enriched foods as part of an overall healthy pattern. ([Perdomo et al., 2020](#)) ([Perdomo et al., 2019](#)) ([Moszak et al., 2021](#)) ([Cinque et al., 2022](#))
- **Coffee and antioxidant-rich foods**
- Some reviews note that unsweetened coffee is a reasonable dietary addition for NAFLD, and antioxidant-rich foods and drinks such as nuts, seeds, vegetable oils, fruits, vegetables, coffee, and tea may help

support liver health. ([Roeb et al., 2021](#)) ([Malhotra et al., 2015](#)) ([Ferri et al., 2023](#)) ([Ivancovsky-Wajcman et al., 2019](#))

- Older and newer reviews also point to vitamins C and E and other natural antioxidants from foods as useful parts of the diet rather than focusing on a single product. ([Kargulewicz et al., 2014](#)) ([Ferri et al., 2023](#))
- Simple plate idea
- A practical way to apply this is: fill half the plate with non-starchy vegetables, add a fiber-rich starch such as beans or whole grains, include fish, poultry, legumes, or eggs for protein, and use olive oil, nuts, or seeds for fat. This matches the food patterns most consistently recommended across the provided papers. ([Perdomo et al., 2020](#)) ([Roeb et al., 2021](#)) ([Zeng et al., 2024](#))

Evidence

([Kargulewicz et al., 2014](#))

[Dietary recommendations for patients with nonalcoholic fatty liver disease](#)

"The diet should be low in calories and rich in unsaturated fatty acids and natural antioxidants (vitamins A and C). According to the nutritional recommendations for patients with NAFLD, carbohydrates should comprise 40-50% of total dietary energy. It is advisable to increase the amount of complex carbohydrates rich in dietary fibre"

"Fat intake should comprise < 30% of daily calory intake. It is essential to increase consumption of food products rich in mono-and polyunsaturated fatty acids"

"Food products recommended for patients with NAFLD should be low in saturated fatty acids [6,(Alwis et al., 2008)[28]. It is advisable to eliminate soft drinks rich in a high-fructose corn syrup (Loria et al., 2010). Meat should be low in fat. Recommended meats are white meat and occasionally red meat. Food should be boiled, steamed or baked"

"Frying should be avoided"

"Using cream or roux is contraindicated. Fast food should be excluded [6]"

"The diet should be rich in fruit and vegetables, which are good sources of antioxidative vitamins (β -carotene, vitamins C and E)."

([Roeb et al., 2021](#))

[Fructose and Non-Alcoholic Steatohepatitis](#)

"intensive lifestyle modifications such as drinking of unsweetened coffee and avoiding fructose corn syrup are reasonable specific dietary recommendation for patients suffering from NAFLD (Malhotra et al., 2015). Mediterranean diets high in vegetables, fish, nuts, grains, fruits, and olive oil enriched with unsaturated fats have been shown to beneficially effect or negatively correlate with NAFLD. In contrast, consumption of a typical Western diet, which includes soft drinks, fructose, red meat and saturated fatty acids predispose for NAFLD development (Zelber-Sagi et al., 2017)"

".five key dietary recommendations including 1) traditional dietary patterns such as the Mediterranean diet, 2) limiting of excess fructose consumption and processed foods and beverages enriched in fructose, 3) replacement of dietary saturated fatty acids by long-chain omega-3 rich foods and monounsaturated fatty acids, 4) substitution of processed food, fast food, commercial bakery goods, and sweets by unprocessed or minimally processed foods that are high in fiber, including whole grains, vegetables, fruits, legumes, nuts, and seeds, and finally 5) reduction of excess alcohol consumption (George et al., 2018)."

([Lv et al., 2024](#))

[Management of non-alcoholic fatty liver disease: Lifestyle changes](#)

"Additionally, diets containing specific macronutrients are good options, including low-carbohydrate and Mediterranean diets (Med diet, MD) (Tosti et al., 2018).An MD is characterized by daily consumption of fresh vegetables, fruits, legumes, minimally processed whole grains, fish (rich in omega-3 fatty acids), olive oil, nuts, and seeds, which are the primary sources of fat (Tosti et al., 2018).As a primary source of fat, minimize or avoid dairy products, red meat, and processed meats."

([Zeng et al., 2024](#))

[The role of dietary modification in the prevention and management of metabolic dysfunction-associated fatty liver disease: an international multidisciplinary expert consensus.](#)

"The recommended dietary strategies emphasize adherence to a balanced diet with controlled energy intake and personalized nutritional interventions, such as calorie restriction, high-protein, or low-carbohydrate diets. Specific dietary advice encouraged increasing the consumption of whole grains, plant-based proteins, fish, seafood, low-fat or fat-free dairy products, liquid plant oils, and deeply colored fruits and vegetables. Concurrently, it advised reducing the intake of red and processed meats, saturated and trans fats, ultra-processed foods, added sugars, and alcohol. Additionally, maintaining the Mediterranean or DASH diet, minimizing sedentary behavior, and engaging in regular physical activity are recommended."

(Perdomo et al., 2020)

Nonalcoholic fatty liver disease and the risk of metabolic comorbidities: how to manage in clinical practice.

"Dietary advice should include caloric restriction and adherence to the macronutrient composition according to the Mediterranean diet [5] (Perdomo et al., 2019)"

"In NAFLD, the carbohydrate intake should include whole grains, unprocessed cereals, and low glycemic index foods (Perdomo et al., 2019); the fat intake should aim at high monounsaturated fatty acids (MUFAs) and omega-3 polyunsaturated fatty acids (PUFAs) consumption; the protein intake should favor vegetable protein, seafood, egg, and white meat consumption. The intake of prebiotic fiber and probiotic enriched products may be recommended to promote a reduced calorie intake and a favorable microbiota, respectively (Perdomo et al., 2019)"

"The Western pattern diet, which is generally characterized by high consumption of carbohydrates, simple sugars, saturated fats, trans fat, animal protein (red meat), processed food, and low fiber intake, is associated with NAFLD development and progression (Oddy et al., 2013)."

(Perdomo et al., 2019)

Impact of Nutritional Changes on Nonalcoholic Fatty Liver Disease

"Non-alcoholic fatty liver disease (NAFLD) is a major global health threat due to its growing incidence and prevalence. It is becoming the leading cause of liver disease in addition to its strong association with cardio-metabolic disease. Therefore, its prevention and treatment are of strong public interest. Therapeutic approaches emphasize lifestyle modifications including physical activity and the adoption of healthy eating habits that intend to mainly control body weight and cardio-metabolic risk factors associated with the metabolic syndrome. Lifestyle interventions may be reinforced by pharmacological treatment in advanced stages, though there is still no registered drug for the specific treatment of NAFLD. The purpose of this review is to assess the evidence available regarding the impact of dietary recommendations against NAFLD, highlighting the effect of macronutrient diet composition and dietary patterns in the management of NAFLD."

(Cinque et al., 2022)

Interaction between Lifestyle Changes and PNPLA3 Genotype in NAFLD Patients during the COVID-19 Lockdown

"The most recommended diet regimen in patients with NAFLD is the Mediterranean diet (MD) (Ryan et al., 2013)[13](Ma et al., 2018) (Gepner et al., 2017)[16], characterized by reduced daily intake of carbohydrates (30% of the whole daily caloric intake, preferably whole grain) and high fat consumption (35-45% of the total energy intake), mainly mono-unsaturated fatty acids (MUFAs) and poly-unsaturated fatty acids (PUFAs)-coming from olive oil, nuts, and fish -with a low intake of saturated fatty acids (<8% of total daily calories) (Godos et al., 2017). Sugary drinks and high fructose corn syrup (HFCS) content foods are forbidden; conversely, increased amounts of fibers found in vegetables, legumes, and whole grains are recommended [18]."

(Ferri et al., 2023)

Effects of COVID-19 Pandemic on Metabolic Status and Psychological Correlates of a Cohort of Italian NAFLD Outpatients

"The most recommended dietary regimen for NAFLD is based on the Mediterranean diet (Ryan et al., 2013)[6]"

"In particular, Mediterranean diet is based on a great reduction in the intake of highly processed food (such as sweets, sugars, sugary drinks, processed meat, ready and pre-cooked foods) in favour of complex carbohydrates (>40% of daily calories, mainly from whole grains, legumes and fibres), fats (35-45% of the total energy, mainly from mono and poly-unsaturated fatty acids) and proteins from vegetal and animal sources, with a preference for dairy products, eggs, fish and white meat. Adequate intakes of vitamins E, C and phenolic compounds, contained in nuts, seeds, vegetable oils, fruits, vegetables, coffee and tea, can protect the liver from NAFLD-related damage and the consumption of up to one glass of wine per day, as a part of a Mediterranean diet, may be considered in patients with non-advanced NAFLD (Clínica, 1971)(Ferri et al., 2022)(Ivancovsky-Wajcman et al., 2019)."

(Tsamis et al., 2023)

Therapeutic Potentials of Reducing Liver Fat in Non-Alcoholic Fatty Liver Disease: Close Association with Type 2 Diabetes

"macronutrients such as saturated fatty acids (SFA), trans fats, simple sugars such as sucrose and fructose, and animal proteins are known to inflict damage on the liver through the accumulation of triglycerides and impaired antioxidant activity, compromising insulin sensitivity and postprandial triglyceride metabolism (Musso et al., 2003). In contrast, the consumption of monounsaturated fatty acids (MUFA), ω 3 polyunsaturated fatty acids (PUFA), plant-based proteins, and dietary fibers such as whole grain cereals,

fruits and vegetables, fatty fish (which are primarily rich in $\omega 3$), and extra virgin olive oil have been found to confer beneficial effects [57,58]. Gupta et al. suggest that oily fish (2-4 g/d), coffee (≥ 3 cups/day), and nuts (100 g/d) are recommended as suitable additions to physical activity and caloric restriction for patients with fatty liver disease, based on strong evidence from human trials. Although tea, red wine, avocado and olive oil can be consumed moderately without harm, more research is needed to investigate their therapeutic benefits for patients with NAFLD/NASH [59]."

(Kasper et al., 2020)

NAFLD and cardiovascular diseases: a clinical review

"Patients with NAFLD should restrict foods high in added fructose, saturated fats, simple carbohydrates and sugar containing drinks while substituting these foods with a low-to-moderate fat and moderate-to-high carbohydrate diet in addition to increasing fruits and vegetables intake (3440607)(Ismaiel et al., 2019). The macronutrient composition should be adjusted according to a Mediterranean diet, which is rich in monounsaturated fatty acids and/or diets rich in long-chain n-3 fatty acids eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) which differ in their effects on cell membrane structure, rates of lipid oxidation, inflammation, and endothelial function (Lechner et al., 2019)(Mason et al., 2020)."

(Ahmed, 2015)

Non-alcoholic fatty liver disease in 2015.

"Diet rich in fructose particularly high fructose corn syrup (Granola bars, condiments, sweetened beverages, prepared desserts, baked goods, snacks, breakfast cereal, cookies) may impair insulin sensitivity leading to development of NAFLD (Ouyang et al., 2008). Thus sugar consumption should be less than 10% of one's total caloric intake and food rich in high fructose corn syrup should be avoided. Western diet is also rich in saturated fat and omega-6 fatty acid but deficient in omega-3 fatty acid (Simopoulos, 2013)"

".Cooking oils high in omega-6 fatty acid (soybean, sunflower, corn) should be changed to cooking oils high in omega-3 fatty acids (Canola, Olive, Chia, Perilla). Patients should be encouraged to eat more fish as they contain omega-3 fatty acid."

(Singh et al., 2017)

Treatment options for alcoholic and non-alcoholic fatty liver disease: A review

"Dietary modification also plays a key role since a carbohydrate-rich diet, especially with high fructose, is the major cause of obesity, insulin resistance and NAFLD development (Ouyang et al., 2008). Thus, sugar consumption should be kept below 10% of total caloric intake in a day and a fructose-rich diet should be avoided in such patients. Food rich in omega-3 fatty acid should be included and those rich in saturated fat and omega-6 fatty acid should be excluded from the diet [142]. An omega-3 fatty acid rich diet promotes fatty acid oxidation and decreases fatty acid synthesis, thus improving the lipid profile. Fish and fish oil consumption should be promoted as they are rich in omega-3 fatty acid (Al-Gayyar et al., 2012)."

(Rinaldi et al., 2021)

Mechanisms of Non-Alcoholic Fatty Liver Disease in the Metabolic Syndrome. A Narrative Review

"A healthy diet refers to the right balance between quantity and quality of food (Vilar-Gomez et al., 2015)(Ahn et al., 2019). Beyond limiting the caloric intake, a correct balance between micro-and macro-nutrients is mandatory. As well, the reduction of the amount of saturated fats and carbohydrates such as fructose is equally important as the intake of lean proteins, fibers and polyunsaturated fatty acids (PUFA) (Francia et al., 2016)(Anania et al., 2018)."

(Sharma et al., 2023)

Uncovering the Cardiovascular Threat: A Comprehensive Examination of Liver Fibrosis and Subclinical Atherosclerosis in Non-alcoholic Fatty Liver Disease

"A balanced diet should emphasize consuming fruits, vegetables, whole grains, lean proteins (such as poultry, fish, and legumes), and foods low in saturated and trans fats. This dietary approach can address multiple aspects of health, including lipid profiles, glycemic control, and liver fat accumulation (Mirmiran et al., 2017)"

".For individuals with NAFLD, it is imperative to abstain from alcohol consumption, as even moderate alcohol intake can exacerbate liver damage. Eliminating alcohol is essential to prevent further harm to the liver and support liver regeneration (Osna et al., 2017)."

(Moszak et al., 2021)

Nutritional Approach Targeting Gut Microbiota in NAFLD—To Date

"Studies recommend avoiding simple sugars (especially fructose) and processed red meat and to increase dietary fiber intake and unsaturated fatty acids. The daily diet should also include lipid-lowering, insulin sensitivity-improving, and anti-inflammatory bioactive substances, as proven in recent years (Hsu et al., 2017)(Saeed et al., 2019)."

(Malhotra et al., 2015)

Management of non-alcoholic fatty liver disease in 2015.

"There is no single pharmacologic therapy that has been approved to treat nonalcoholic fatty liver disease in the general population. The backbone of therapy currently includes intensive lifestyle modification with established targets for diet and weight loss. The use of unsweetened, unfiltered coffee along with limiting high fructose corn syrup have emerged as beneficial dietary recommendations. The use of empiric oral hypoglycemic agents and vitamin E, however, has not been widely accepted. Developing bariatric surgical techniques are promising, but additional studies with long-term follow up are needed before it can be widely recommended. Finally, liver transplantation is an increasingly frequent consideration once complications of end-stage disease have developed. The future treatment of those with nonalcoholic fatty liver disease will likely involve a personalized approach. The importance of the gut microbiome in mediating hepatocyte inflammation and intestinal permeability is emerging and may offer avenues for novel treatment. The study of anti-fibrotic agents such as pentoxifylline and FXR agonists hold promise and new pathways, such as hepatocyte cannabinoid receptor antagonists are being studied. With the incidence of obesity and the metabolic syndrome increasing throughout the developed world, the future will continue to focus on finding novel agents and new applications of existing therapies to help prevent and to mediate the progression of nonalcoholic fatty liver disease."

(Ivancovsky-Wajcman et al., 2019)

Dietary vitamin E and C intake is inversely associated with the severity of nonalcoholic fatty liver disease.

"BACKGROUND & AIMS

Although antioxidants have a protective potential in nonalcoholic fatty liver disease (NAFLD) and steatohepatitis (NASH), there is limited evidence regarding the role of dietary intake of antioxidants. The aim was to test the association between dietary vitamins E and C intake and NAFLD, NASH and fibrosis markers.

METHODS

Cross-sectional study of a large cohort of subjects undergoing colonoscopy. The presence of NAFLD was evaluated by ultrasonography. The level of steatosis was defined using SteatoTest, moderate-severe NASH using new quantitative NashTest and borderline-significant fibrosis $\geq$ F12 using FibroTest. Nutritional intake was measured by food frequency questionnaire (FFQ).

RESULTS

Overall, 789 subjects were included (52.6% men, age 58.83 ± 6.58 years), 14 had reliable FibroMax. Adjusting for BMI, dietary and lifestyle factors, the upper tertile of vitamin E intake/1000 Kcal was associated with lower odds of NASH (OR = 0.60, 95% CI 0.43-0.94, $P = 0.024$). There was an inverse association between reaching the recommended vitamin E intake and NASH (OR = 0.48, 95% CI 0.30-0.77, $P = 0.002$). The upper tertile of vitamin C intake/1000 Kcal was associated with lower odds of NAFLD and NASH (OR = 0.68, 95% CI 0.47-0.99, $P = 0.045$; OR = 0.50, 95% CI 0.38-0.84, $P = 0.004$, respectively). Both vitamins were related with the level of steatosis according to SteatoTest.

CONCLUSION

Vitamin E and C intake may be protective from NAFLD-related liver damage."

Foods, beverages, and exposures to avoid or minimize

Cut back most on sugar-sweetened drinks, sweets, refined carbs, fast food, and foods high in saturated fat. Also limit red and processed meat, sodium, and highly processed foods, and avoid rapid weight loss from overly aggressive dieting. (7 sources)

- **Sugar-sweetened drinks**
- This is one of the clearest "avoid" categories across the papers. Reviews advise avoiding or strongly discouraging sweetened drinks and soda because excess sugar, especially fructose from beverages, is linked with higher triglycerides, weight gain, NAFLD risk, and fibrosis progression. ([Mirmiran et al., 2017](#)) ([Ongel et al., 2023](#)) ([Fuente, 2022](#)) ([Yoneda et al., 2023](#))
- **Sweets, desserts, and added sugars**
- Minimize sweets, sugary desserts, and foods with a lot of added sugar. These are repeatedly grouped with the foods patients with fatty liver disease should reduce because they raise simple carbohydrate and fructose intake. ([Fuente, 2022](#)) ([Ongel et al., 2023](#)) ([Mirmiran et al., 2017](#))
- **Refined carbohydrates and simple carbs**

- Cut back on white bread, pastries, sugary breakfast foods, and other refined starches and simple carbohydrates. Several papers specifically advise avoiding simple carbohydrates or reducing refined carbohydrate intake as part of NAFLD care. ([Mirmiran et al., 2017](#)) ([Fuente, 2022](#))
- **Fast food and ultra-processed foods**
- Fast food should be minimized. This fits with the broader message to reduce highly processed foods, which often combine refined carbs, unhealthy fats, salt, and excess calories. ([Fuente, 2022](#))
- One review also highlights concern about oxidized cholesterol compounds such as 7-ketocholesterol, which can be present in processed foods and foods affected by freezing and cooking, and which has been linked with hepatic steatosis in experimental work. This does not mean patients must fear every cooked food, but it adds to the case for eating fewer heavily processed foods. ([Yoneda et al., 2023](#)) ([Chang et al., 2021](#))
- **Saturated fat**
- Saturated fat is another main target to reduce. The papers repeatedly advise avoiding or lowering saturated fats, which usually means eating less butter, cream, high-fat cheese, fatty cuts of meat, and many commercial baked or fried foods. ([Mirmiran et al., 2017](#)) ([Yoneda et al., 2023](#))
- **Red and processed meat**
- Limit red meat and especially processed meat. These are listed among the foods to reduce in Mediterranean-style eating patterns recommended for NAFLD. ([Fuente, 2022](#))
- **High sodium foods**
- Salt intake should be limited, even early in disease. This is recommended both for overall cardiometabolic health and because higher sodium intake has been associated with greater NAFLD risk in observational research. In practice, the biggest sodium sources are often packaged snacks, fast food, processed meats, canned soups, and restaurant meals rather than table salt alone. ([Ongel et al., 2023](#)) ([Jamiol-Milc et al., 2023](#)) ([Bischoff et al., 2020](#))
- **Excess fructose from sweetened beverages and processed foods**
- Beyond “sugar” in general, the papers specifically call out excess fructose intake from beverages and products made with fructose-dextrose, high-fructose liquid sugar, or isomerized sugar. Patients do not need to avoid whole fruit based on these quotes; the concern is mainly concentrated added fructose in drinks and processed foods. ([Yoneda et al., 2023](#))
- **Rapid weight-loss diets**
- Avoid crash dieting or trying to lose weight too fast. One paper recommends keeping weight loss below about 1.6 kg per week to avoid worsening liver cell injury and fibrosis risk, so steady, sustainable loss is preferred over extreme restriction. ([Ongel et al., 2023](#))
- **Alcohol**
- If fatty liver disease is specifically alcohol-related, alcohol should be avoided; if it is nonalcoholic fatty liver disease, patients should ask their clinician how much alcohol, if any, is safe because guidance varies by liver disease stage and individual risk. (Model-Generated)

Evidence

(Mirmiran et al., 2017)

[Relationship between Diet and Non-alcoholic Fatty Liver Disease: A Review Article](#)

"all patients with NAFLD are advised to adhere to several dietary recommendations such as avoiding simple carbohydrates, saturated fats, and sweetened drinks and consuming diets rich in fruits and vegetables (Nseir et al., 2014)(Sofi et al., 2014) (Liangpunsakul et al., 2003)."

(Ongel et al., 2023)

Thermodynamic Assessment of the Effects of Intermittent Fasting and Fatty Liver Disease Diets on Longevity

"For obese patients, 5-10% weight loss is recommended, but protein intake must be monitored to prevent muscle mass loss (Clínica, 1971). To avoid hepatocyte necrosis (death of liver cells) and fibrosis from worsening, weight loss is recommended to be less than 1.6 kg per week (Andersen et al., 1991)"

"An energy intake of 35-40 kcal/kg per day and a protein intake of 1.2-1.5 g/kg per day is recommended for NAFLD patients (Plauth et al., 2006). Including late-night snacks and breakfast in the diet might be beneficial in meeting the daily caloric needs (Clínica, 1971). In EASL (Clínica, 1971) and ESPEN (Plauth et al., 2006) guidelines, the importance of branched-chain amino acid (BCAA) consumption is highlighted; therefore, the majority of the protein intake should come from dairy protein sources. If the patient cannot uptake enough calories, leucine-enriched BCAA supplements might be used. The consumption of sugared beverages and desserts is strongly discouraged; in addition, sodium intake should be limited as well (Plauth et al., 2006). In the nutrition of liver patients, zinc and vitamin E supplementation were found highly beneficial (Hanje et al., 2006)."

(Fuente, 2022)

Nutrition and Chronic Liver Disease

"All in all, fast food should be minimised and it is preferable to follow a Mediterranean diet (Zelber-Sagi et al., 2016)"

"The Mediterranean dietary pattern is most often recommended. It is based on the consumption of olive oil, nuts, oily fish, legumes, fruits, vegetables and predominantly monounsaturated and polyunsaturated fats (omega-3). Conversely, consumption of soda drinks, sweets, red and processed meats, and refined carbohydrates should be reduced to only 40% of total kilocalories (Salas-Salvad et al., 2014)(Ryan et al., 2013)(Nordmann et al., 2011)(Opie, 2013)."

(Yoneda et al., 2023)

Nonalcoholic Fatty Liver Disease as a Systemic Disease and the Need for Multidisciplinary Care

"Conversely, n-3 unsaturated fatty acids increase insulin sensitivity and decrease intrahepatic fat mass. The EASL-EASD-EASO clinical practice guidelines therefore recommend the Mediterranean diet, which is rich in tri-saturated fatty acids and low in carbohydrates, for patients with NAFLD. (Romero-Gomez et al., 2017)"

"7-ketocholesterol, an oxidized form of cholesterol produced by freezing and cooking of foods and present in processed foods, has been associated with the development of CVD and hepatic steatosis. (Song et al., 2017)(Chang et al., 2021)"

"Excessive fructose intake is associated with increased blood triglycerides, weight gain, increased risk of NAFLD, and progression of liver fibrosis; however, the target fructose intake in NAFLD is unknown. However, fructose intake from beverages and foods containing fructose dextrose, high fructose liquid sugar, and isomerized sugar has recently increased."

(Chang et al., 2021)

Dietary Oxysterol, 7-Ketocholesterol Accelerates Hepatic Lipid Accumulation and Macrophage Infiltration in Obese Mice

"Non-alcoholic fatty liver disease is strongly associated with obese and type 2 diabetes. It has been reported that an oxidized cholesterol, 7-ketocholesterol (7KC), might cause inflammatory response in macrophages and plasma 7KC concentration were higher in patients with cardiovascular diseases or diabetes. Therefore, we have decided to test whether small amount of 7KC in diet might induce hepatic steatosis and inflammation in two types of obese models. We found that addition of 0.01% 7KC either in chow diet (CD, regular chow diet with 1% cholesterol) or western type diet (WD, high fat diet with 1% cholesterol) accelerated hepatic neutral lipid accumulation by Oil Red O staining. Importantly, by lipid extraction analysis, it has been recognized that triglyceride rather than cholesterol species was significantly accumulated in CD+7KC compared to CD as well as in WD+7KC compared to WD. Immunostaining revealed that macrophages infiltration was increased in CD+7KC compared to CD, and also in WD+7KC compared to WD. These phenotypes were accompanied by inducing inflammatory response and downregulating fatty acid oxidation. Furthermore, RNA sequence analysis demonstrated that 7KC reduced expression of genes which related to autophagy process. Levels of LC3-II protein were decreased in WD+7KC compared to WD. Similarly, we have confirmed the effect of 7KC on acceleration of steatohepatitis in db/db mice model. Collectively, our study has demonstrated that small amount of dietary 7KC contributed to accelerate hepatic steatosis and inflammation in obese mice models."

(Jamiol-Milc et al., 2023)

Nutritional Support for Liver Diseases

"the grade B recommendation points out the Mediterranean diet as the recommended diet for NAFLD patients, which reduces steatosis and increases insulin sensitivity"

"Additionally, salt intake should be limited already in the earliest stages of the disease for the purposes of the prevention of hypertension and cardiovascular diseases. In addition, a systematic review and meta-analysis of observational studies showed that populations with a high sodium intake have a 60% greater risk of developing NAFLD compared to those with low consumption, but the explanation of the mechanism of this phenomenon requires further research [43]."

(Bischoff et al., 2020)

[ESPEN practical guideline: Clinical nutrition in liver disease.](#)

"BACKGROUND

The Practical guideline is based on the current scientific ESPEN guideline on Clinical Nutrition in Liver Disease.

METHODS

It has been shortened and transformed into flow charts for easier use in clinical practice. The guideline is dedicated to all professionals including physicians, dietitians, nutritionists and nurses working with patients with chronic liver disease.

RESULTS

A total of 103 statements and recommendations are presented with short commentaries for the nutritional and metabolic management of patients with (i) acute liver failure, (ii) alcoholic steatohepatitis, (iii) non-alcoholic fatty liver disease, (iv) liver cirrhosis, and (v) liver surgery/transplantation. The disease-related recommendations are preceded by general recommendations on the diagnostics of nutritional status in liver patients and on liver complications associated with medical nutrition.

CONCLUSION

This practical guideline gives guidance to health care providers involved in the management of liver disease to offer optimal nutritional care."

Caloric restriction, macronutrient targets, and practical implementation

For most patients with fatty liver disease, the practical goal is a moderate calorie deficit that leads to slow, steady weight loss, not a crash diet. A useful template is a Mediterranean-style diet with controlled portions, moderate carbs, enough protein, and lower saturated fat. (6 sources)

In practical terms, the diet usually works best when calories are reduced enough to produce gradual weight loss. The papers provided repeatedly recommend a deficit of about 500 to 1000 kcal/day, which usually aims for roughly 0.5 to 1.0 kg weight loss per week. (Paul, 2020) (Coilly et al., 2016) Some papers also frame this as reducing usual intake by about 600 to 800 kcal/day or targeting about 25 to 30 kcal/kg/day of ideal body weight. (Arab et al., 2014) In guideline-style summaries, example hypocaloric intake ranges are about 1000 to 1200 kcal/day for women and 1200 to 1500 kcal/day for men, though individual needs still depend on body size, age, activity, and medical context. (Coilly et al., 2016) (Nascimbeni et al., 2013)

For macronutrients, the broad message is to control total calories first, then shape those calories toward better quality. The cited papers suggest keeping **carbohydrates around 40% to 45% of calories**, with some reviews recommending **less than 50%**, while avoiding excess fructose and refined carbohydrate. (Arab et al., 2014) (Paul, 2020) Protein intake is often suggested at about 1 to 1.5 g/kg/day, which can help preserve lean mass during weight loss. (Arab et al., 2014) Fat intake is generally shifted away from saturated fat and toward unsaturated fat: one review summarizes targets of **<30% of calories from fat** and **<10% from saturated fat**, while another emphasizes lowering cholesterol, saturated fat, and fructose and increasing **n-3 polyunsaturated fats**. (Arab et al., 2014) (Paul, 2020) (Coilly et al., 2016)

Diet quality still matters even when weight loss is the main goal. Evidence summarized in these papers suggests that diets lower in fat, saturated fat, and glycemic load can improve liver fat and insulin sensitivity, even without major weight change, while overfeeding saturated fat appears to drive more liver fat accumulation than polyunsaturated fat. (Coilly et al., 2016) (Utzschneider et al., 2012) (Rosqvist et al., 2014) So the practical implementation is not just "eat less," but "eat less of the foods that worsen liver fat, and more of the foods that improve overall diet quality."

A simple way to apply this day to day is to set up meals with **portion control**, keep sugary drinks and sweets out of the routine, and use a repeatable meal pattern rather than trying to count every nutrient perfectly. Patients can start by shrinking portions of calorie-dense foods, using vegetables and legumes to add fullness, choosing whole-grain or other high-fiber carbs in modest portions, and including a protein source at meals. This approach fits the calorie-deficit targets above while staying close to the Mediterranean-style pattern recommended across the papers. [\(Paul, 2020\)](#) [\(Arab et al., 2014\)](#) [\(Coilly et al., 2016\)](#)

If patients want a simple bottom line: aim for **steady weight loss, not rapid loss**; use a **moderate calorie deficit**; keep **carbs moderate and mostly high-fiber**; get **adequate protein**; and keep **saturated fat, fructose, and excess calories** low. That is the most practical diet framework supported by the references provided. [\(Paul, 2020\)](#) [\(Arab et al., 2014\)](#) [\(Coilly et al., 2016\)](#)

Evidence

(Paul, 2020)

[Recent advances in non-invasive diagnosis and medical management of non-alcoholic fatty liver disease in adult](#)

"Diet modification: Energy restriction with a low calorie, low fat, and low carbohydrate or high-protein Mediterranean diet is advised in NAFLD patients. 500-1000 kcal deficit diet per day to induce a weight loss of 500-1000 g/week is usually recommended (according to EASL, Asia-Pacific and AASLD guidelines). According to the Italian Association for the Study of the Liver (IASL), amount of fat and carbohydrate should be < 30% and < 50% of total calories respectively"

". Fructose-containing beverages and foods should be avoided because it appears as one major factor for initiation of hepatic steatosis and also its progression to NASH and more severe stages of the liver fibrosis [103,104]. Alcohol intake should be < 30 g/ day for men and < 20 g/ day for women but better to avoid alcohol consumption of any type or amount in individuals with NAFLD [105]."

(Coilly et al., 2016)

[Recent Insights into Treatment of Non-Alcoholic Steatohepatitis](#)

"The obvious goal would be to modify the diet by decreasing cholesterol, Saturated Fatty Acid (SFA) and fructose intake, and increase n-3 Polyunsaturated Fatty Acid (PUFA) intake. The whole caloric intake has to be decreased by 25% in calories from the normal diet based on patient's sex and age according to the World Gastroenterology Organization Global Guidelines (Utzschneider et al., 2012)(Rosqvist et al., 2014)[29][30]"

".It is now recommended that hypocaloric diet should provide 1000-1200 calories per day for women and 1200-1500 calories per day for men with a goal to achieve a weight loss of 0.5-1.0 kg per week based on published guidelines (Nascimbeni et al., 2013)"

".Furthermore, low-fat diet associated with weight loss program improves body weight, Body Mass Index (BMI), fatty liver, insulin sensitivity and plasma triglycerides [36]. Importantly, heavy alcohol consumption should be avoided in NAFLD patients [37]. Mediterranean diet due to preparation with olive oil, rich in n-3 PUFA, might be of additional help to modify hepatic lipid content and decrease cardiovascular risks [38]."

(Arab et al., 2014)

[Management of nonalcoholic fatty liver disease: an evidence-based clinical practice review.](#)

"The dietary recommendations for patients with NAFLD should include a reduction of 600-800 calories in the usual daily oral intake or caloric restriction set to 25-30 kcal/kg per day of the ideal body weight (Ueno et al., 1997)(Palmer et al., 1990)(Vajro et al., 2004). Protein intake should be 1-1.5 g/kg per day, and carbohydrates should be restricted to 40%-45% of the caloric intake (Huang et al., 2005). Of the total daily fat in the diet, saturated fat should also be restricted to < 30% of the caloric intake; < 10% of the caloric intake should be saturated fat (Hickman et al., 2004)(Kugelmas et al., 2003)(Schäfer et al., 2007). The patient should be instructed to preferentially consume fruits and vegetables instead of fructose-rich food (Huang et al., 2005)"

"High-cholesterol/high-fat diets, with low levels of polyunsaturated fat, fiber and antioxidants (vitamins C and E), are associated with NASH (Vajro et al., 2004)(Kugelmas et al., 2003)(Musso et al., 2003)."

(Nascimbeni et al., 2013)

[From NAFLD in clinical practice to answers from guidelines.](#)

"This review of the literature consists of three sections. First, papers concerning non-alcoholic fatty liver disease (NAFLD) awareness among the general population, general practitioners, and liver and non-liver specialists were retrieved and analyzed to highlight the perception of disease, verify knowledge of current recommendations, and identify the main difficulties experienced in

clinical practice. Next, position papers and clinical practice guidelines issued by International and National Hepatological Scientific Societies were identified and critically assessed in order to pinpoint the areas of convergence/difference. Finally, practical suggestions on NAFLD diagnosis and management in daily practice are provided and the open questions highlighted."

(Utzschneider et al., 2012)

Beneficial effect of a weight-stable, low-fat/low-saturated fat/low-glycaemic index diet to reduce liver fat in older subjects

"Non-alcoholic fatty liver disease is associated with insulin resistance and dyslipidaemia and can progress to steatohepatitis and cirrhosis. We sought to determine whether dietary fat and saturated fat content alter liver fat in the absence of weight change in an older population. Liver fat was quantified by magnetic resonance spectroscopy before and after 4 weeks on an isoenergetic low-fat/low-saturated fat/low-glycaemic index (LGI) (LSAT: 23 % fat/7 % saturated fat/GI < 55) or a high-fat/high-saturated fat/high-GI (HSAT: 43 % fat/24 % saturated fat/GI > 70) diet in older subjects. In the present study, twenty subjects (seven males/thirteen females; age 69.3 (sem 1.6) years, BMI 26.9 (sem 0.8) kg/m²) were randomised to the LSAT diet and fifteen subjects (six males/nine females; age 68.6 (sem 1.8) years, BMI 28.1 (sem 0.9) kg/m²) to the HSAT diet. Weight remained stable. Liver fat decreased significantly on the LSAT diet (median 2.2 (interquartile range (IQR) 3.1) to 1.7 (IQR 1.8) %, $P = 0.002$) but did not change on the HSAT diet (median 1.2 (IQR 4.1) to 1.6 (IQR 3.9) %). The LSAT diet lowered fasting glucose and total cholesterol, HDL-cholesterol and LDL-cholesterol and raised TAG ($P < 0.05$), while the HSAT diet had no effect on glucose or HDL-cholesterol but increased total cholesterol and LDL-cholesterol ($P < 0.05$). Fasting insulin and homeostasis model of insulin resistance did not change significantly on either diet, but the Matsuda index of insulin sensitivity improved on the LSAT diet ($P < 0.05$). Assignment to the LSAT v. HSAT diet was a predictor of changes in lipid parameters but not liver fat. We conclude that diet composition may be an important factor in the accumulation of liver fat, with a low-fat/low-saturated fat/LGI diet being beneficial."

(Rosqvist et al., 2014)

Overfeeding Polyunsaturated and Saturated Fat Causes Distinct Effects on Liver and Visceral Fat Accumulation in Humans

"Excess ectopic fat storage is linked to type 2 diabetes. The importance of dietary fat composition for ectopic fat storage in humans is unknown. We investigated liver fat accumulation and body composition during overfeeding saturated fatty acids (SFAs) or polyunsaturated fatty acids (PUFAs). LIPOGAIN was a double-blind, parallel-group, randomized trial. Thirty-nine young and normal-weight individuals were overfed muffins high in SFAs (palm oil) or n-6 PUFAs (sunflower oil) for 7 weeks. Liver fat, visceral adipose tissue (VAT), abdominal subcutaneous adipose tissue (SAT), total adipose tissue, pancreatic fat, and lean tissue were assessed by magnetic resonance imaging. Transcriptomics were performed in SAT. Both groups gained similar weight. SFAs, however, markedly increased liver fat compared with PUFAs and caused a twofold larger increase in VAT than PUFAs. Conversely, PUFAs caused a nearly threefold larger increase in lean tissue than SFAs. Increase in liver fat directly correlated with changes in plasma SFAs and inversely with PUFAs. Genes involved in regulating energy dissipation, insulin resistance, body composition, and fat-cell differentiation in SAT were differentially regulated between diets, and associated with increased PUFAs in SAT. In conclusion, overeating SFAs promotes hepatic and visceral fat storage, whereas excess energy from PUFAs may instead promote lean tissue in healthy humans."